

Project Management

CSE 4407

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The Spark: Origins of Systems Projects

- Project Triggers
 - Problems
 - Opportunities

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 - **Opportunities:** Chance to improve, upgrade, or innovate. New tech, changing markets, e-commerce potential
- **Driving Forces**
 - Adapting to organizational change (growth, new strategies)
 - Responding to external shifts (legal, industry, e-commerce trends)
- **Key Takeaway:** Many ideas are suggested, but only some become formal projects after evaluation

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 - Listen to External Feedback
 - From: Vendors, Customers, Suppliers, Service Providers
 - Via: Complaints, Suggestions for improvement, Loss of sales, Lower-than-expected sales

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 - **Constraints:** Limitations or boundaries (Budget, deadlines, technology restriction - often include "not")

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 - **Primary:** It’s the very first thing the user brings up

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- **Validation:** Analyst should try to witness the problem firsthand if possible

Case Study: Catherine's Catering Conundrum

- **The Business:** Small catering company (meals, receptions, banquets). Started small, good reputation led to growth

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- **The Decision:** Hired IT/business consultants to help

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 - (Summary Reports) Provide queries/reports for summary info

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- **Next Step:** These requirements drive the creation/modification of Use Cases or Data Flow Diagram (more on these later!)

Catherine's Catering: Thinking Ahead - Testing

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- Catherine's Preliminary Test Plan
 - Test viewing all product types
 - Test submitting requests (valid and invalid data) → correct routing
 - Test adding clients (validation, correct credentials)
 - Test client event viewing and update functionality (including the 5-day rule)
 - Test HR system (adding employees, scheduling logic, constraints)
 - Test reports queries for accuracy

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- **Goal:** Select projects that provide genuine value and align with the organization's direction

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3. Improves Strategic Organizational Goals?

- Does it align with the big picture? E.g., Improve profits? Support competitive strategy? Enhance vendor/partner cooperation? Improve internal operations (efficiency)? Improve internal decision-making? Improve customer service? Boost employee morale?

Project Go/No-Go: Practicality and Worth (2/2)

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- 4. Practical in Terms of Resources?

- Do *we* (analysts/dev team) have the necessary skills and tools?
 - Does the *organization* have the capacity (staff, infrastructure)?
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- Is this the *best* use of limited resources right now? (Opportunity Cost)
 - Compare against other potential projects or improvements, such as Speeding up processes, Streamlining (removing steps), Combining processes, Reducing input errors (better forms/screens), Moving systems to the cloud, Reducing redundant storage/output, Improving system integration, etc.

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- **Analogy:** Checking if you have the time, budget, and necessary items *before* planning a trip

Can We *Really* Do This? The T.E.O. Test

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 - Whether the system will operate when put in service
 - Whether the system will be used

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- Focus: Availability and capability of technology and skills

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- Do the expected *benefits* outweigh the costs?
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 - Business employee time (training, transition)
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- **If Not Economically Viable:** Stop the project

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 - Lack of user involvement in requesting the system
 - Poor user interface design (Covered later in Ch. 14)

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 - Lack of user involvement in requesting the system
 - Poor user interface design (Covered later in Ch. 14)
- **Positive Signs**
 - User themselves requested the change
 - Users see clear benefits (efficiency, accessibility, reliability)

Feasibility Deep Dive: Operational Check

- **Assumes:** Tech is possible, Economics make sense
- **Key Questions**
 - Will the system *operate correctly* within the organization's environment?
 - Will people *actually use* the system once it's deployed?
 - How will it impact workflows and processes?
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- **Focus:** Human resources, organizational culture, process integration

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- Example Outcome
 - Proposed system significantly reduces human and computer time
 - Supports economic and operational feasibility arguments

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 - Inventory computer hardware
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- **Goal:** Make informed decisions about acquiring the right tools for the job

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- **Link to Staffing:** Helps assess if current staff skills match existing/needed hardware

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- **Important:** Define required/desired functions *before* vendor demos!

Own It or Rent It? Buy vs. Cloud Hardware

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- **Hybrid Approach:** Often organizations use a mix (some owned, some cloud)

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- Focus [1]
 - Primarily IaaS for hardware needs
 - SaaS later for software
 - PaaS is in between

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- **Don't Forget**
 - Check vendor stability
 - Read the fine print (SLAs, contracts) → involve legal if needed!

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- **Major Drawback**
 - SECURITY RISK! (Lost/stolen devices, malware, insecure Wi-Fi usage, unauthorized access)
 - Requires strong security policies and management tools (MDMs)

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- **Reality:** Many systems use a mix! (e.g., COTS for accounting, Custom for core process, SaaS for CRM)

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- **Remember:** Vendors certify software works, but don't guarantee it's error-free in all situations or compatible with everything else you run. *Test thoroughly!*

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- Interdependence: Costs and Benefits must be considered *together*
- Forecasting is Key
 - To analyze costs/benefits over time, we need to *predict* key variables (e.g., future usage volume, labor costs, sales)
 - Relies on historical data or judgment methods (if no data)

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- **Example:** Moving Average → Average change of data series over time
 - Smooths out fluctuations to reveal trends
 - Calculates average over a fixed period (e.g., 3-month average predicts month 4)
 - Simple, but sensitive to extreme values

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- **Intangible Benefits:** Difficult to quantify, but still important!

- Improved decision-making process
 - Enhanced data accuracy
 - Improved customer service
 - Reputation
 - Increased employee job satisfaction

What's the Upside? Identifying Benefits

- Two Types of Benefits
 - **Tangible Benefits:** Measurable in dollars
 - Increased processing speed → Reduced labor time
 - Access to new information → Better decisions leading to profit/savings
 - Timelier information → Faster response to opportunities/threats
 - Superior calculation power → Complex analysis possible
 - Decreased employee time on tasks → Labor cost savings
 - Reduced errors → Lower correction costs
 - **Intangible Benefits:** Difficult to quantify, but still important!
 - Improved decision-making process
 - Enhanced data accuracy
 - Improved customer service
 - Reputation
 - Increased employee job satisfaction
- **Crucial:** Include BOTH tangible and intangible benefits in the proposal for a complete picture

What's the Downside? Identifying Costs

- Two Types of Costs (Parallel to Benefits)
 - Tangible Costs
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Technique 1: Break-Even Analysis

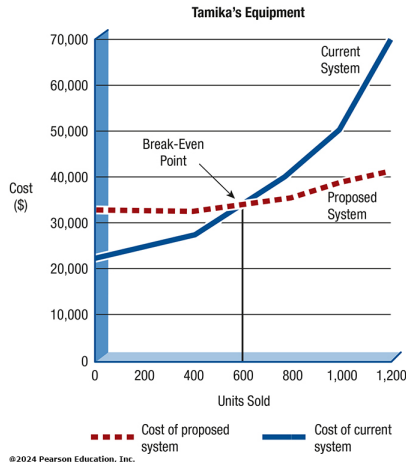
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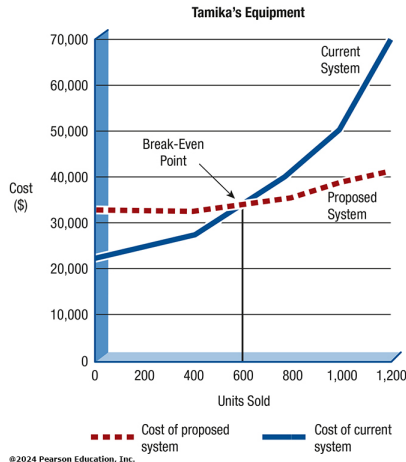
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- Beyond this point, the proposed system is cheaper *per unit of volume*



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Year	Cost (\$)	Cu. Costs (\$)	Ben. (\$)	Cu. Ben. (\$)
0	30,000	30,000	0	0
1	1,000	31,000	12,000	12,000
2	2,000	33,000	12,000	24,000
3	2,000	35,000	8,000	32,000
4	3,000	38,000	8,000	40,000
5	4,000	42,000	10,000	50,000
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- **Payback Analysis**
 - **Purpose:** Determines how long it takes for the accumulated *benefits* (especially tangible ones) of the new system to “pay back” the initial development and setup costs
- **Combined View:** Often, both are used alongside other financial metrics for a full economic picture

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- **Completeness**
 - All tasks must add up to 100% of the project work
 - Tasks can vary in duration and team size

Ways to Structure the WBS

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- Choice Depends On

- Project type
 - How you want to manage/track progress
 - Process-oriented is common in SAD

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- **Five Common Techniques**
 1. Relying on Experience
 2. Using Analogies
 3. Three-Point Estimation
 4. Function Point Analysis
 5. Using Time Estimation Software

Estimation Techniques: Drawing From The Past

- **Relying on Experience**

- Best approach if you have done the same tasks before
- Provides estimates based on real-world knowledge (including potential pitfalls)
- Gives “most likely” and “pessimistic” estimates

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- **Using Analogies**

- Used when direct experience is lacking, but you've done *something* similar
- Identify a past project (even unrelated) with comparable structure/complexity
- Compare the WBS/Network diagrams of both projects
- Base estimates for the new project on the known durations from the analogous one (adjusting for differences)

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- **Weighted Average:** $E = \frac{a+4 \times m+b}{6}$
- **Example:** Coding a module
 - Best: 8 days
 - Most likely: 10 days
 - Worst: 30 days
 - $E = \frac{8+4 \times 10+30}{6} = 13$ days
- **Result:** Provides a more realistic single estimate than just using 'm'

Estimation Techniques: Specialized

- **Function Point Analysis (FPA)**

- Estimates effort based on the system's *functional size* and complexity, NOT lines of code initially
- Measures five components: External Inputs, External Outputs, External Queries, Internal Logical Files, External Interface Files
- Complexity ratings are applied
- Can be used to compare estimated effort across different Programming languages
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- **Using Time Estimation Software**

- Based on models like COCOMO II or COSYSMO (e.g., SystemStar software)
- Analyst inputs estimated system size (e.g., lines of code, function points) AND other factors (team experience, platform, required reliability, etc.)
- Software calculates rough estimates for effort, duration, staffing
- Estimates become more refined as the project progresses

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- **Foundation:** A detailed WBS

Phase	Activity
Analysis	Data gathering Data flow and decision analysis Proposal preparation
Design	Data entry design Input design Output design Data organization
Implementation	Implementation Evaluation

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- **Foundation:** A detailed WBS
- **Key**
 - Detail must be sufficient for scheduling and control
 - Time estimates are added

Phase	Activity	Detailed Activity	Wk.
Analysis	Data gathering	Conduct interviews	3
		Administer questionnaires	4
		Read company reports	4
		Introduce prototype	5
		Observe reactions to prototype	3
	Data flow and decision analysis	Analyze data flow	8
Design	Proposal preparation	Perform cost-benefit analysis	3
		Prepare proposal	2
		Present proposal	2
	Data entry design	-	-
		-	-
Implementation	Input design	-	-
		-	-
	Output design	-	-
		-	-
Implementation	Data organization	-	-
		-	-
Implementation	Implementation	-	-
		-	-
Implementation	Evaluation	-	-
		-	-

Visualizing the Time: Gantt Charts

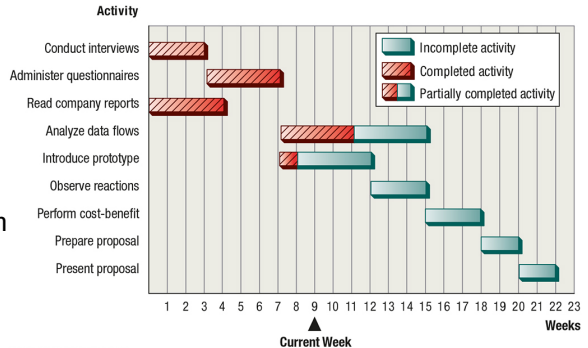
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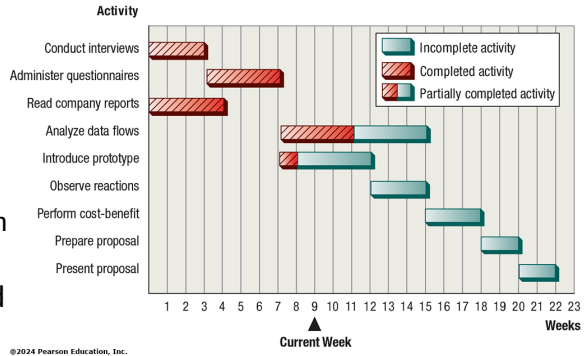
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- **Advantages:** Simple, easy to create and understand, good for communicating progress visually



Gantt Chart Limits and A More Powerful Tool: PERT

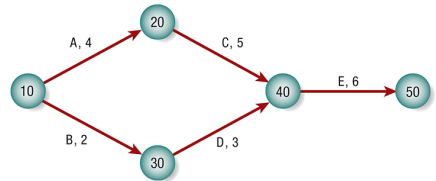
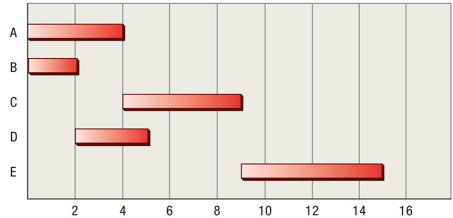
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- **Enter PERT:** Program Evaluation and Review Technique
 - Developed for complex projects (US Navy Polaris)
 - A network diagram showing tasks and their dependencies
 - Excellent for projects where tasks can happen **in parallel**
 - Helps identify the **critical path**

Visualizing Dependencies: Gantt vs. PERT

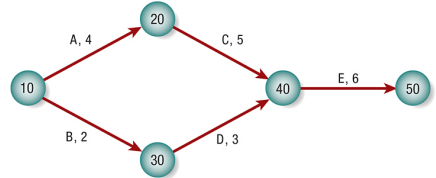
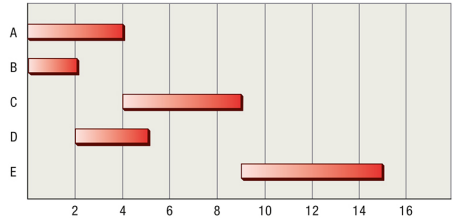
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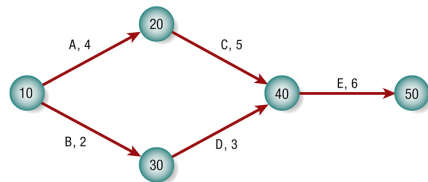
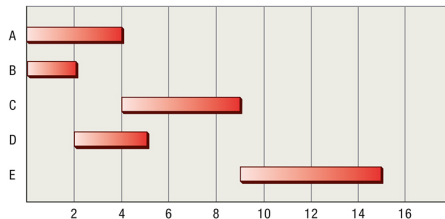
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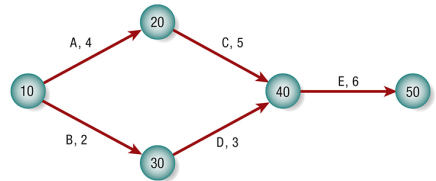
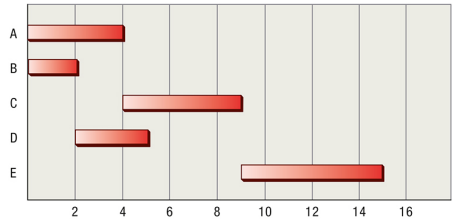
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- **Key Difference:** PERT explicitly shows precedence
 - C can't start until A is done at node 20
 - E can't start until *both* C and D are done



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PERT Concepts: Critical Path

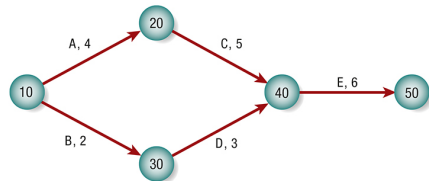
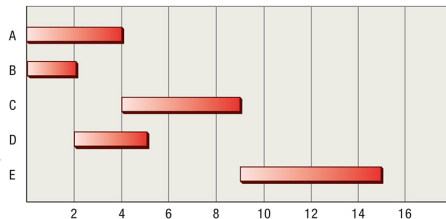
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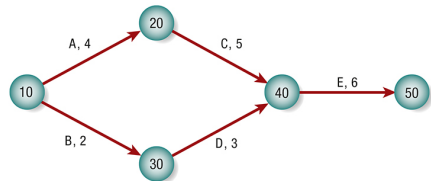
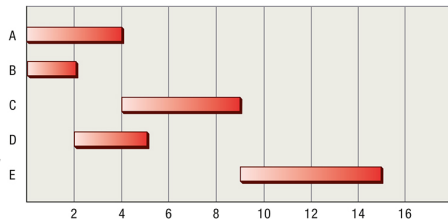
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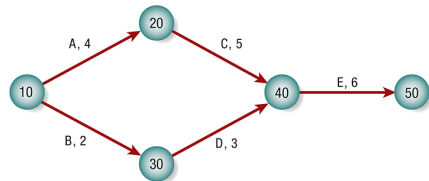
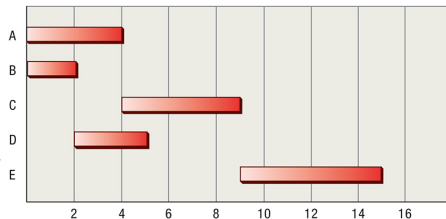
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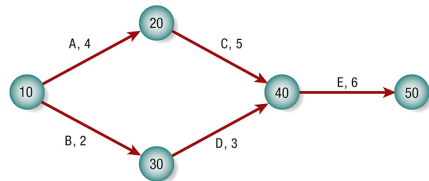
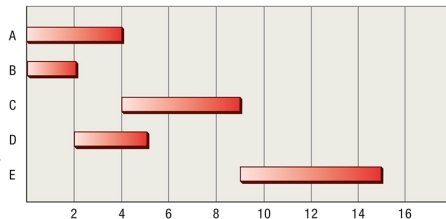
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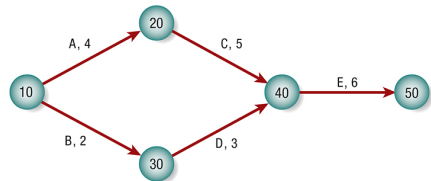
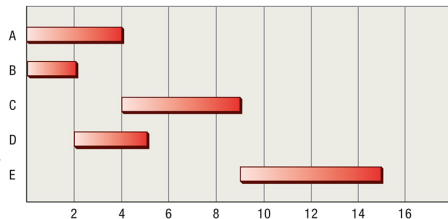
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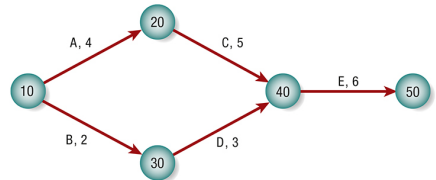
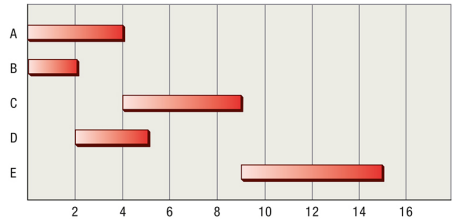
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PERT Concepts: Slack Time

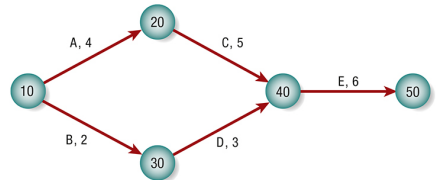
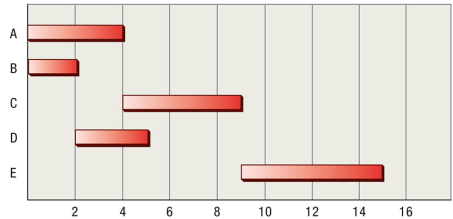
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- Exists only on non-critical paths



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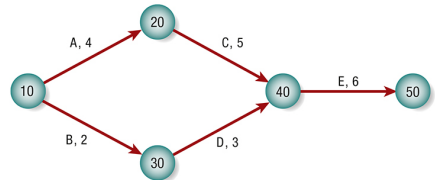
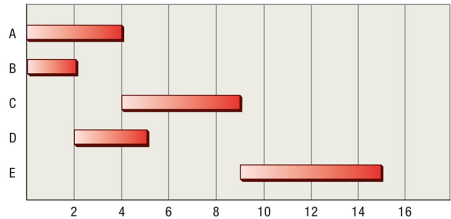
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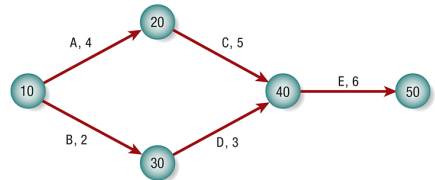
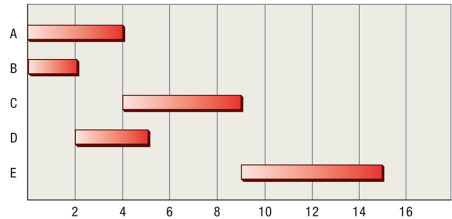
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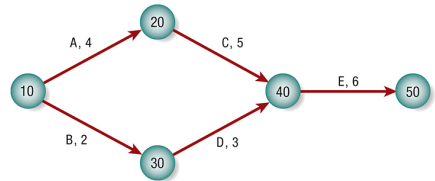
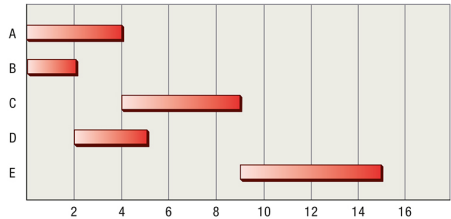
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 - Implies Tasks B or D could slip by a total of 4 weeks without impacting the project duration



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- Exists only on non-critical paths
- Slack Time for a Path = Project Duration - Non-critical Path Duration
- **Example**
 - Slack on Path 2 = Project Duration - Path 2 Duration = 4 weeks
 - Implies Tasks B or D could slip by a total of 4 weeks without impacting the project duration
 - Also implies Tasks A, C, and E have zero slack



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PERT Concept: Dummy Activities

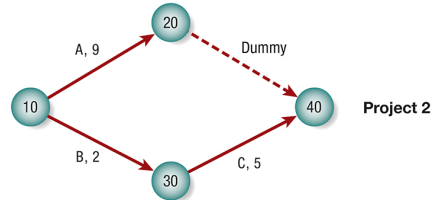
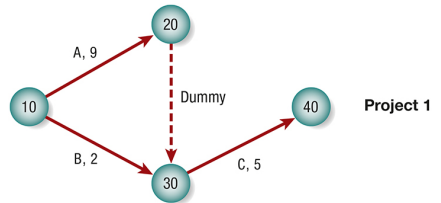
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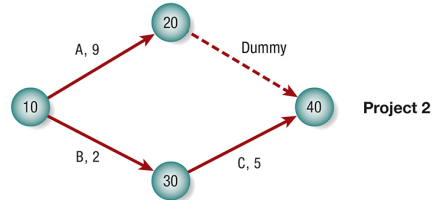
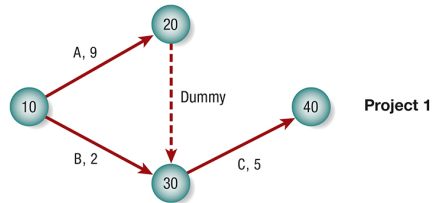
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- **Purpose:** Used to maintain correct logic and precedence relationships, especially when tasks share *some* but not *all* predecessors
- **Key:** Dummies clarify precedence when standard arrows alone would create incorrect logic



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PERT Example: Data Gathering and Proposal

- **Scenario:** Scheduling the initial phases of a systems project

Activity		Predecessor(s)	Duration (Weeks)
A	Conduct interviews	None	3
B	Administer questionnaires	A	4
C	Read company reports	None	4
D	Analyze data flow	B, C	8
E	Introduce prototype	B, C	5
F	Observe reactions prototype	E	3
G	Perform cost-benefit	D	3
H	Prepare proposal	F, G	2
I	Present proposal	H	2

PERT Example: Constructing the Network

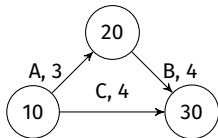
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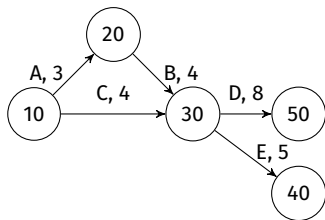
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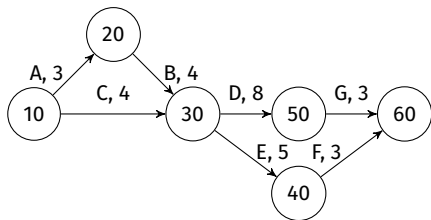
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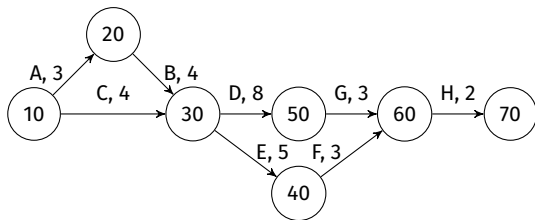
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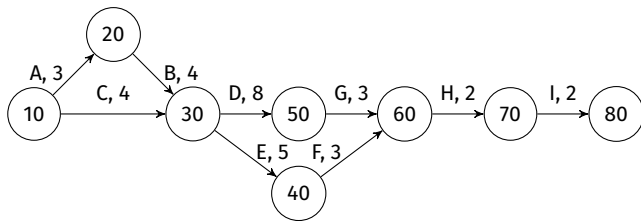
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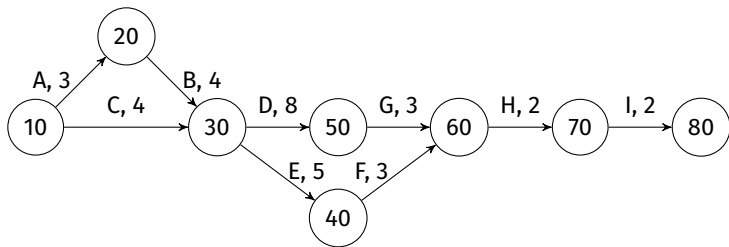
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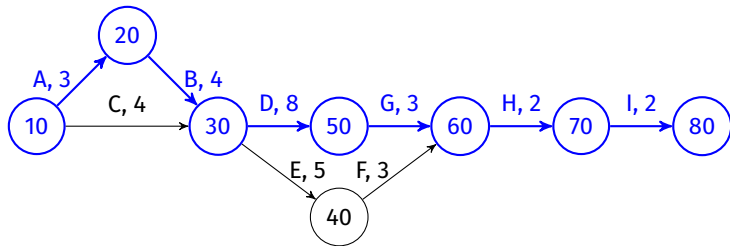
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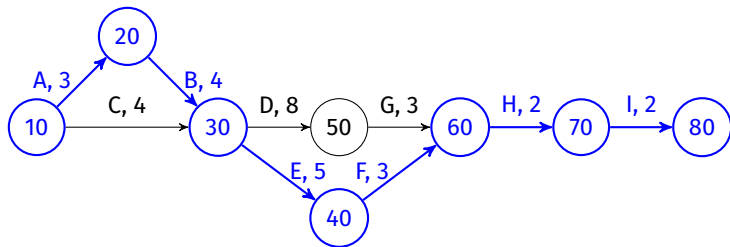
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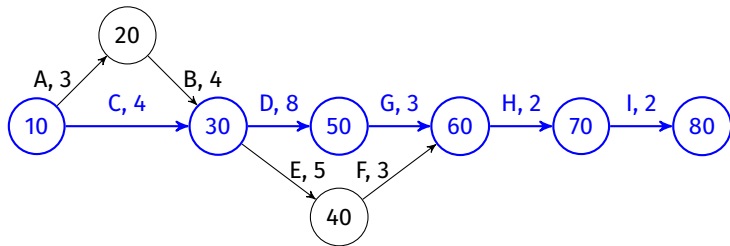
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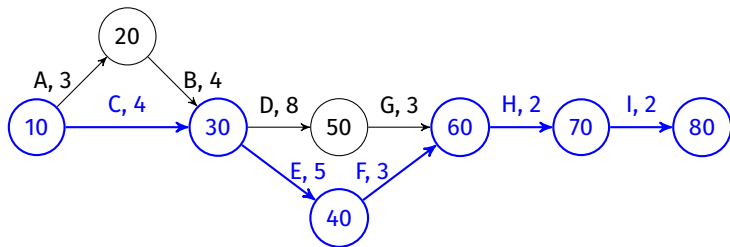
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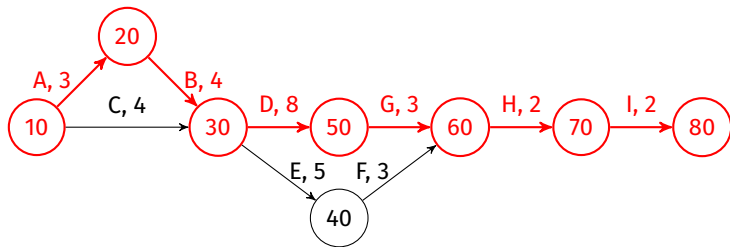
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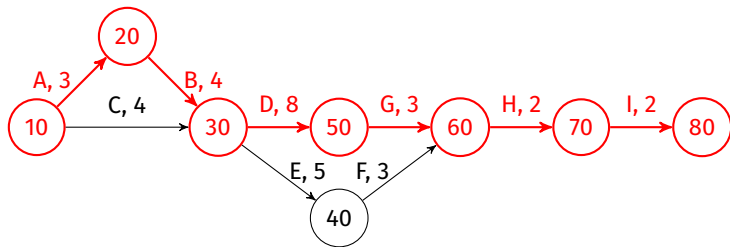
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- Management Focus:** Activities, A, B, D, G, H, and I must be carefully monitored. Activities C, E, and F have some slack.

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- **Key Control Areas:** Cost, Risk, Time

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- **Common Practice:** Use a combination of methods

Controlling Costs: Pitfalls and Budgeting

- Why Cost Estimates Often Fail
 - Over-Optimism
 - Rushing

Demo: [Sample Budget](#)

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- Key: Be as accurate as possible, knowing estimates *will* be revised
- Preparing the Budget: A critical project deliverable!
 - Clients/Management need it early
 - Often uses standard organizational forms/templates
 - Details costs by category (Team time, HW, SW, Training, etc.)

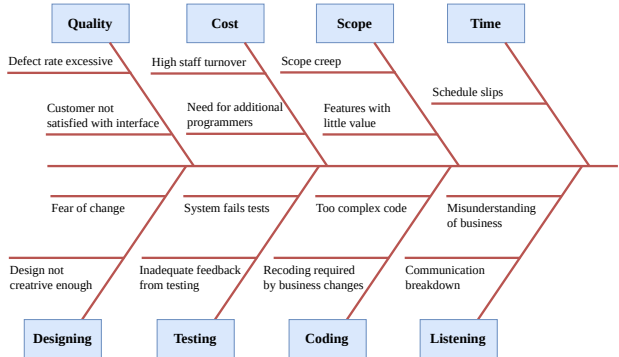
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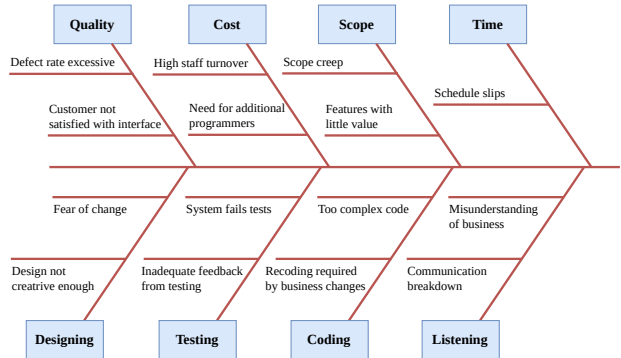
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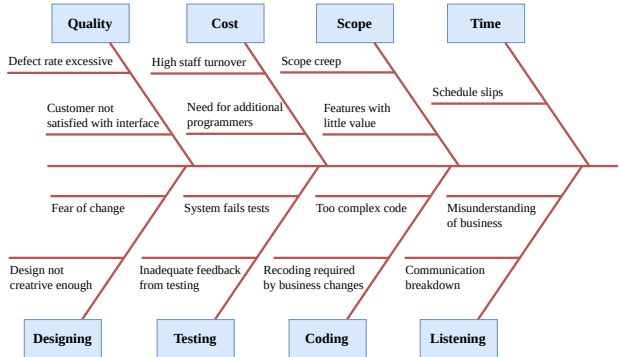
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- **Remember:** Management has the final say, but team's reputation is linked to project success



Need for Speed? Expediting Activities

- **Crashing:** Speeding up project activities to finish earlier
- **Why**
 - Potential bonus for early completion
 - Free up resources/team members for other projects sooner
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Activity	Estimated Duration	Crash Time	Cost/Week
A	3	1	\$800
B	4	2	500
C	4	2	400
D	8	6	1,000
E	5	5	1,000
F	3	3	800
G	3	3	800
H	2	2	400
I	2	1	600

How to Expedite: The Analysis

- **Rule #1:** Only expedite critical-path activities
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Project Duration: 22 weeks → 15 weeks

Cost: \$5,600

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- **PMBOK definition:** $PV = BAC \times \text{Expected } p\%$ and $EV = BAC \times \text{Actual } p\%$

EVM in Action: Website Example

Project Budget: \$18,000

At the End of	Stage	Estimated Cost	Cu. Estimate	Estimated Duration	Stage Completed	Actual Cost of Stage to Date	Actual Cost of Project to Date
Month 1	1	\$6,000	\$6,000	1 month	100%	\$6,000	\$6,000
Month 2	2	3,000	9,000	1 month	100%	3,000	9,000
Month 3	3	3,000	12,000	1 month	100%	3,000	12,000
Month 4	4	3,000	15,000	1 month	50%	5,000	17,000
Month 5	5	3,000	18,000	1 month	0%	Not yet begun	Not yet begun

- EVM at End of Month 4
 - $BAC = \$18,000$
 - $PV = \$15,000$ (Cumulative Estimated Cost)
 - $AC = \$17,000$
 - $EV = ?$ (Needs to be calculated)

EVM Calculations: Where Do We Stand?

After 4 months,

- Calculate % Work done so far (p): $\frac{100\%+100\%+100\%+50\%}{100\%+100\%+100\%+100\%} = \frac{350}{400} = 0.875$

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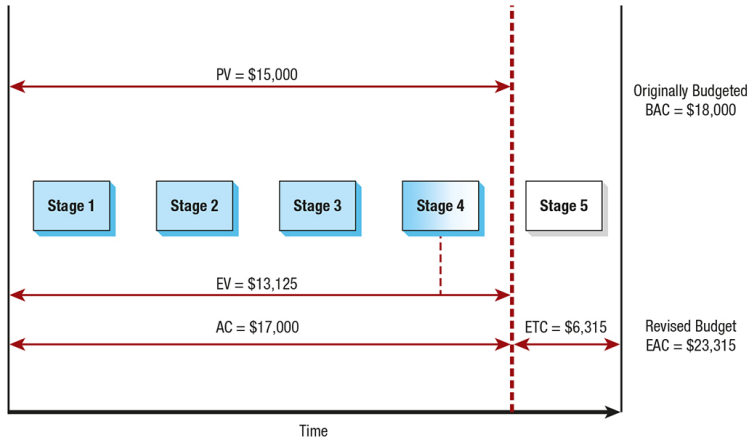
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- **Website Example:** $ETC = \frac{\$18,000 - \$13,125}{0.772} \approx \$6,315$ and $EAC = \$17,000 + \$6,315 = \$23,315$
- **Result:** Project likely to cost ~\$23,315 instead of the planned \$18,000

EVM Visualization: Key Takeaway

- **Analyst's Role:** Balance Cost, Time, and Scope based on this information



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It's About People: Managing the Project Team

- **Equally Important:** Managing the people doing the work

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- **Key Aspects**
 - Assembling the right team
 - Fostering effective communication and dynamics
 - Setting achievable productivity goals
 - Motivating team members
 - Understanding unique project contexts (e.g., E-commerce)
 - Formalizing expectations (Project Charter)

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- **Diversity and Inclusion Matters**
 - **Why:** Diverse teams outperform individuals, make faster/better decisions, solve problems faster (cognitive diversity) [3], [4]
 - **How:** Fair hiring, pay equity, support for individual success, ensuring every voice is heard [5]

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- Also Valuable

- **Experience:** Especially for time/cost estimation and avoiding pitfalls (Experienced devs can be much faster)
- **Enthusiasm and Imagination:** Drives innovation and problem-solving
- **Communication Skills:** Strong writers/speakers for proposals, user interaction, etc.
- **Usability Expert:** Focuses on making the system user-friendly

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- **Managing Tension**
 - Ignoring tension leads to dysfunction
 - Open communication and feedback are key to resolving issues arising from the task/relationship balance

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- **Action:** Teams need to make norms explicit and periodically assess if they are helping or hindering. *Change should be the norm!*

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- **How Goal Setting Motivates**

- **Clarity:** Team knows exactly what is expected
- **Achievement Focus:** Goals act as targets, creating focus
- **Autonomy:** Often defines the “what”, allowing team members’ flexibility in the “how”
- **Feedback:** Performances measured against clear goals simplifies reviews

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 - Scattered Data
 - Cross-department (Marketing, Sales, Inventory, Finance)
 - Increases complexity and dept. politics
 - Diverse Teams
 - Skills: Dev, Marketing, DB, Security, Integration
 - External partners; fluid teams
 - Integration Focus
 - Front-end ↔ Inventory, Billing, Shipping
 - Often the hardest part
 - Heightened Security
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- Management Tips

- Align goals and foster cross-department integration
 - Engage partners early; keep communication clear

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- **Analysis Methods** to be used
- Key **Participants** and Time Commitment
- Project **Deliverables** (Specific outputs)
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- **Result:** Shared understanding, reduced ambiguity, clear definition of 'done'

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- **Audience:** Management, IT Task Force, Key Users

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4. **Executive Summary**

- The “TL;DR” (Who, What, When, Where, Why, How)
- Recommendations and Desired Action (250-375 words)
- Write LAST

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- 8. Systems Analysts' Recommendations

- The team's chosen solution and *why*
 - Must flow logically from alternatives analysis

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- 10. **Appendices**

- Supporting info (detailed data, charts, correspondence, etc.)

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 - Cost-benefit comparison tables
 - Break-even/Payback data
 - Hardware/Software option comparisons

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 - Break-even analysis graph
 - Payback period visualization
 - Comparison of performance benchmarks
 - User satisfaction ratings

References I

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